Communic-aid: Learning How Much Evidence to Ask For in Gaze-Based Messaging on a Commodity iPhone

VANDYA BISARIA, MANSIE SINGH, SINNGAM KHAIDEM, and MANAN SOLANKI, Indian Institute of Technology Gandhinagar, India

Gaze is often the most reliable input left to people with severe motor impairments, but gaze interfaces still force a single trade off on every action: a long dwell is slow, a short dwell selects things by mistake, and a fixed confirmation step is either too weak for a call or too tedious for a letter. We present Communic-aid, a hands-free messaging interface that uses only an iPhone with a TrueDepth camera as its sensor. The phone streams gaze, head pose, blinks, eyebrow raises and a spoken confirmation word to a laptop, which runs a chat app with calls, messages and a gaze keyboard. Our central idea is that the interface should learn how much evidence to ask for before it acts, for each person, each context and each level of risk. Every action belongs to one of four risk tiers with an accident cost. A dwell learner sets the dwell time from implicit feedback (undos, quick backspaces, abandoned looks and stray glances) using Thompson sampling, and a contextual bandit chooses which second signal, if any, to ask for on risky actions. A preliminary study with two able-bodied participants shows that the learner shortened the dwell from 1500 ms to about 1200 ms within one session, with two accidents in 146 selections, both on undoable letters and none on risky actions, that the router adapted its choice of confirmation signal to each participant within a session, and that a grouped keyboard with large targets was 1.5 to 2.8 times faster than QWERTY for gaze typing. We describe the planned within-subject comparison against a fixed baseline, including a participant with a motor impairment.

CCS Concepts: • **Human-centered computing** → **Accessibility technologies**; *Interaction techniques.*

Additional Key Words and Phrases: gaze interaction, adaptive dwell time, multimodal confirmation, contextual bandit, motor impairment, augmentative and alternative communication, commodity hardware

1 Introduction

For many people with ALS, muscular dystrophy, cerebral palsy or spinal cord injury, the eyes remain controllable long after the hands do. Gaze interfaces build on this, but they inherit the Midas touch problem [4]: the eyes are always looking at something, so the interface cannot tell a look that means "do this" from a look that means "I am reading". The standard answer is a dwell time. A long dwell is safe and slow; a short dwell is fast and causes accidental selections. Both kinds of error hurt, but not equally. Accidentally typing a letter costs one backspace. Accidentally calling someone or deleting a message costs far more.

Existing systems address parts of this. Adaptive dwell methods adjust the dwell to the typist [6, 10, 12], and multimodal methods add a second channel to confirm a selection [3, 11]. Our literature review found three gaps that these systems leave open. First, confirmation is uniform: low cost and high cost actions face the same bar. Second, adaptive dwell methods mostly learn from explicit signals, and treat a false selection and a missed selection as equally bad. Third, no system we found chooses the confirmation modality itself from experience, even though people differ greatly in which signals they can produce reliably [14, 19]. Specialised eye trackers also remain costly and obtrusive, which our interviewees and prior work identify as a barrier [9, 20].

We present Communic-aid, which treats these as one problem: *how much evidence should the interface ask for before it acts?* It answers per person, per context and per level of risk, and it learns the answer from how the person actually uses it. Our contributions are:

Authors' Contact Information: Vandya Bisaria; Mansie Singh; Sinngam Khaidem; Manan Solanki, Indian Institute of Technology Gandhinagar, Gandhinagar, India.

- A hands-free messaging system whose only sensor is an iPhone with a TrueDepth camera, combining gaze or head pointing with four confirmation signals: blink and hold, raised eyebrows, a spoken “go”, and an optional switch.
- Risk tiers that give every action an accident cost, so the strength of confirmation scales with what an error would cost.
- A dwell learner that sets the dwell time per tier and context from implicit feedback alone, with asymmetric steps that lengthen quickly after accidents and shorten slowly.
- A contextual bandit that ranks confirmation modalities from past interactions and asks for the one most likely to succeed at the lowest effort.
- Preliminary results from two able-bodied participants, and a study plan for the comparison against a fixed baseline.

2 Related Work

Dwell and the Midas touch. Dwell selection is the default for gaze input, and the right dwell depends on the person and the task [13]. Majaranta et al. let typists adjust the dwell themselves and saw speed rise with practice [6]. Mott et al. varied the dwell per key with the probability of the next letter [10]. GazeIntent adapts the dwell with a learned model of intent and found that most returning users preferred the personalised version [12], and Meena et al. adapt a multimodal keyboard over time [8]. These methods tune one threshold for all actions, and most learn from explicit corrections. Communic-aid learns from implicit outcomes and weighs accidents by the cost of the action.

Second signals. Pairing gaze with another channel reduces accidental selections: silent speech for people with ALS [3], gestures paired with dwell on mobile devices [11], and anomaly detection that stops likely errors [14]. Facial gestures can be personalised to the user [7], but recognisers show accuracy gaps for people with motor impairments [19]. This argues against fixing one confirmation signal in advance, and for choosing it from what works for each person, in the spirit of ability-based design [18].

Commodity hardware. GazeSpeak showed that a smartphone camera can support gaze gesture communication for people with motor disabilities [20], and webcam keyboards bring gaze typing to low resource settings [9]. Since 2024 iOS also includes system eye tracking with dwell control [1]. Communic-aid uses the phone differently: as a sensor that streams gaze, head pose, facial blendshapes and speech to a larger screen, so that one device provides both pointing and several confirmation signals.

3 Formative Input

We interviewed three people with motor impairments (muscular dystrophy, cerebral palsy, and a locomotor disability), using questions adapted from the WebAIM survey [17]. Together with the literature, they shaped five requirements: (1) avoid accidental actions, above all for calls and deletions; (2) adapt to the individual rather than to an average user; (3) use a second modality for confirmation, but sparingly, since too many confirmations are tiring; (4) keep the setup cheap and unobtrusive, with no special hardware; and (5) tolerate involuntary and atypical movement, including speech that dictation handles poorly. One interviewee already sent voice notes instead of long texts because typing took too long, which led us to treat typing effort as a primary cost.

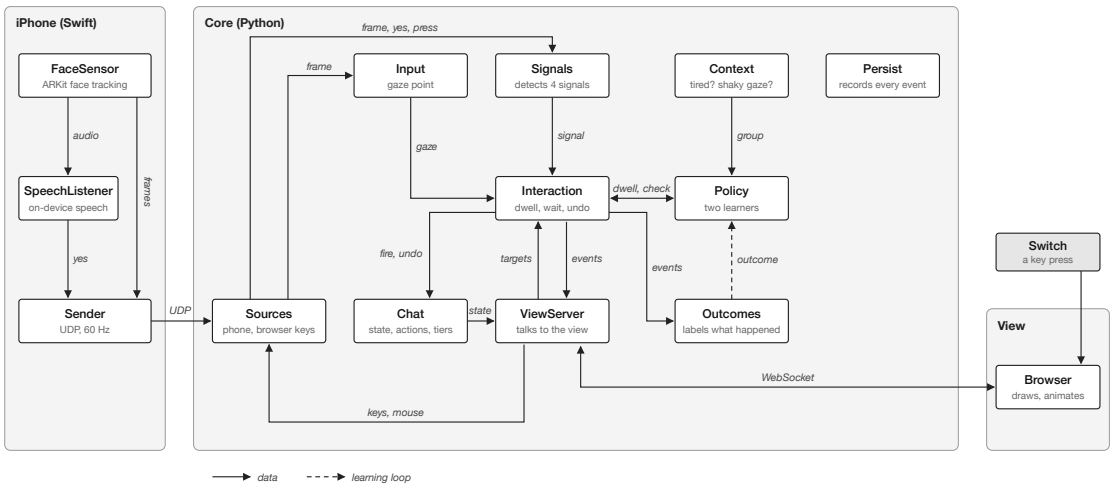


Fig. 1. Architecture. The iPhone only senses. The Python core makes every decision: it maps gaze to the screen, detects signals, runs the dwell and the confirmation step, labels outcomes and updates the two learners. The browser only draws.

Table 1. Risk tiers. The accident cost enters both learners. Tier 0 and 1 actions fire on dwell alone; tier 2 and 3 actions add a confirmation step after a short fixed dwell of 1000 ms.

Tier	Actions	Cost	Confirmation
0	Open a chat, go back, scroll, navigation	1	Learned dwell
1	Letters, word suggestions, Space, Backspace, settings	2	Learned dwell
2	Send, react, end a call	4	The router chooses dwell alone or a signal
3	Call, delete, forward	8	Always a signal, chosen by the router

4 System

4.1 Sensing with one phone

The phone sits below the screen, 40 to 60 cm from the face. An iOS app built on ARKit face tracking sends, at 60 Hz over UDP, the gaze ray and eye position in camera space, the direction of the nose, and the blink and eyebrow blendshapes. Speech recognition runs on the phone and listens only for the word “go”. A Python core on the laptop maps the gaze ray to the screen with a nine point calibration (a quadratic least squares fit with a small ridge term), smooths it with an exponential filter, holds it still during blinks, and corrects slow drift using the centres of buttons the person has just selected. In head pointing mode the same pipeline uses the nose direction instead of the gaze ray. Signals are detected from the blendshapes: a blink held for 500 ms, eyebrows raised for 300 ms, or the spoken word. A one time signal check measures each person’s rest and peak levels and sets each threshold halfway between them.

4.2 Risk tiers

Every action belongs to one of four tiers (Table 1). The cost of an accident doubles with each tier. Undo is always available for 5 seconds through an undo strip on the screen, so the cost of a tier 0 or 1 accident is a short repair, while a call that has started cannot be taken back.

4.3 Labelling outcomes without asking

The learners never ask the user for feedback. Instead the core labels each look and each confirmation request from what happens next. A selection that is undone within 5 s, or a letter that is deleted within 3 s, is an *accident*; otherwise it *worked*. A look that filled at least half of the dwell, ended, and was followed within 5 s by a fresh selection of the same button is a *give-up*: the person wanted it but the dwell was too long. Any other unfilled look is a *stray glance*. For confirmation requests, a signal that arrives is *worked*, a request abandoned after trying is a *give-up*, and a request that times out with no attempt at all is *blocked*: the confirmation step stopped an action the person did not want.

4.4 Context

Each learner keeps separate statistics for four context groups: rested or tired (more than 20 minutes into the session, or in the evening) crossed with steady or shaky gaze (the recent spread of gaze on targets more than 1.5 times a steady baseline). Until a group has enough data of its own, it borrows the pooled data of all groups.

4.5 Dwell learner

For tiers 0 and 1, the dwell learner keeps six counts per group: intended and unintended looks, stray glances and their total duration, and give-ups and the total waiting time. From these it draws, by Thompson sampling [16], a probability π that a look is intended (from a Beta distribution), a mean stray glance length g and a mean patience p (from Gamma distributions). For each candidate dwell d between 800 and 2000 ms it scores

$$V(d) = \pi \left[(1 - G(d))\,(1 - \lambda d) + r_{\text{give-up}}\, G(d) \right] - (1 - \pi)\, A(d)\, c_{\text{tier}},$$

where $A(d) = e^{-d/g}$ is the chance that a stray glance lasts long enough to fire, $G(d) = 1 - e^{-d/p}$ is the chance that the person gives up first, $\lambda = 0.6$ per second is the cost of waiting, $r_{\text{give-up}} = -0.5$, and c_{tier} is the accident cost. The best d is a target, not a setting: after each selection the dwell moves 5% of the way towards it, by at most 10 ms when shortening and up to 100 ms when lengthening, so that accidents are corrected quickly while speed is gained cautiously. The dwell starts at 1500 ms and stays there for the first 20 selections in a group.

4.6 Modality router

For tiers 2 and 3, a contextual bandit [5] chooses the confirmation step. The arms are dwell alone (tier 2 only), blink and hold, raised eyebrows, the word “go”, and a switch. For each arm, tier and group it keeps a count of the four outcomes (worked, give-up, blocked, accident) and draws their probabilities from a Dirichlet distribution. The expected value of an arm weighs the outcomes by their rewards: a worked signal earns 1 minus its effort, which grows with the person’s average time to produce that signal; a blocked accident earns a small positive reward proportional to the tier cost; an accident costs the full tier cost. The arm with the highest sampled value is asked for. The person’s preferred signal, chosen in settings, starts with a small bonus, and a group borrows the pooled counts until it has 20 tries of its own. Because sampling keeps some uncertainty, a signal that worked poorly early on is still tried occasionally rather than being ruled out for good.

4.7 Interface

The view is a full screen chat app with a chat list, threads, calls, message actions and settings. A look fills the target button; a confirmation request shows which signal to give and waits up to 4 seconds. Text entry offers two keyboards: a

Table 2. Free use session. Selections exclude undo and redo. Head movement is the distance of the eyes from their median position; turning is the angle of the nose from its median direction.

Measure	P1	P2
Session length	9 min	24 min
Calibration error	80 px	69 px
Selections (tier 0 / 1 / 2 / 3)	51 (18 / 29 / 2 / 2)	95 (32 / 51 / 7 / 5)
Accidents	2 (both tier 1)	0
Undos	2	0
Dwell give-ups	6	11
Stray glances	145	541
Confirmation requests on tiers 2 and 3	5	13 (4 dwell alone)
worked / gave up / accidents	4 / 1 / 0	12 / 1 / 0
Dwell at end, tier 0 / tier 1	1216 / 1333 ms	1219 / 1431 ms
Head movement, median / 95th percentile	18 / 214 mm	18 / 69 mm
Head turning, median / 95th percentile	4 / 42°	3 / 11°

grouped keyboard with eight large letter groups (abc, def, …, wxyz) where a letter takes two selections, and a QWERTY keyboard with one selection per letter. Word suggestions appear above both.

5 Preliminary Study

5.1 Method

Two able-bodied adults (P1, P2) used the system with gaze pointing in the adaptive condition, on a laptop with the phone mounted below the screen and their heads free to move. After the signal check and calibration, each used the chat app freely and completed the study tasks: navigating between chats and settings, composing and sending a message of at least ten letters, making and ending a call, and deleting a message. P1's session lasted 9 minutes and P2's 24 minutes. Each then did a typing test: copying a 248 character passage for two minutes, without word suggestions, three times with each keyboard, grouped first, with a new passage in every run. We report words per minute (WPM = $(n-1)/t \times 60/5$ for n characters in t seconds), accuracy from the edit distance to the passage, and keystrokes per character [15]. All measures come from the core's event log.

5.2 Results

Accidents and confirmation. Table 2 summarises the free use sessions. Across 146 selections there were two accidents, both on tier 1 letters, and both undone. No tier 2 or 3 action fired by mistake. The router asked for a signal 14 times and for dwell alone 4 times; 16 of the 18 requests succeeded and 2 were given up. The router adapted within each session. When P1 gave up on raised eyebrows to confirm Send, the next Send asked for blink and hold, which worked, as did blink and hold for the call and the deletion that followed. P2 started with blink and hold as the preferred signal; after three successful confirmations, the router began to accept dwell alone on tier 2 actions (Send, React, End call), and all four such actions worked. After P2 changed the preferred signal to "go" in settings, the router asked for it on the remaining risky actions, and P2 confirmed "go" five times out of five.

Dwell learning. Stray glances outnumbered selections by three to six times, which shows how much implicit evidence a gaze interface produces even when nothing is selected. After its 20 selection warm-up the dwell learner shortened both tiers steadily (Figure 2). By the end of the free use session, P1's tier 0 dwell had fallen from 1500 to 1216 ms and P2's to 1219 ms. Tier 1 stayed longer (1333 and 1431 ms), as expected from its higher accident cost and, for P1,

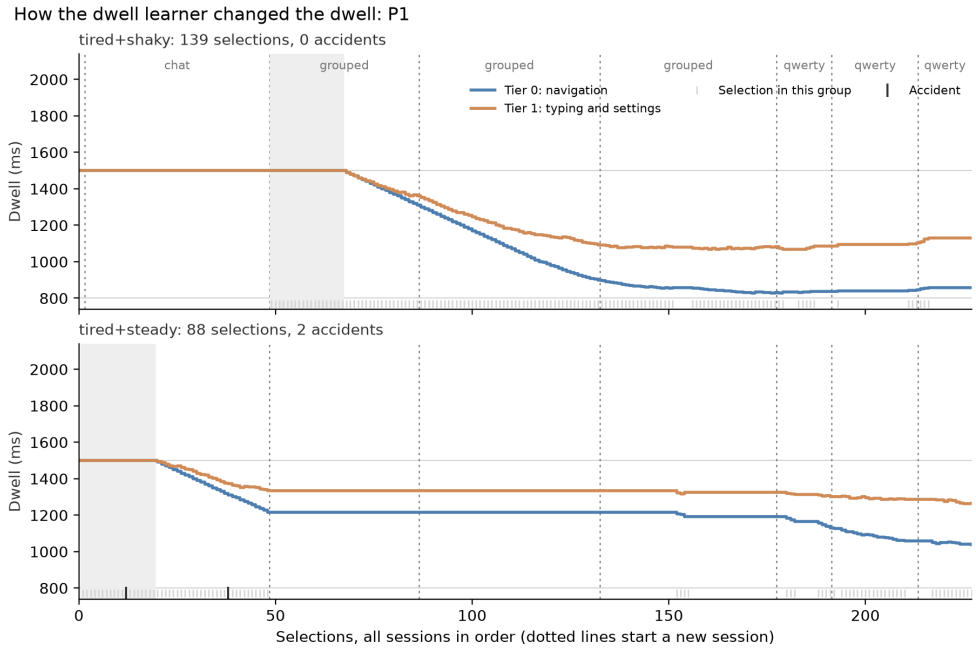


Fig. 2. P1's dwell over all sessions, one panel per context group and one line per tier. The shaded area is the warm-up; the first session is the free use session, followed by three grouped and three QWERTY typing runs. Marks along the bottom are the group's selections (light) and accidents (dark).

Table 3. Typing test, mean of three two minute runs per keyboard. The mean tier 1 dwell in use during the runs is shown because the dwell was still learning during this test.

	Keyboard	WPM	Accuracy	Keystrokes per char.	Mean dwell
P1	Grouped	2.23	100%	1.84	1261 ms
	QWERTY	1.50	91.7%	1.06	1214 ms
P2	Grouped	1.60	100%	1.86	1327 ms
	QWERTY	0.57	100%	1.11	1259 ms

from the two tier 1 accidents. In P1's later typing runs, a second context group (tired with shaky gaze) appeared and started from the 1500 ms default, which the learner then shortened separately to about 860 ms for tier 0 and 1130 ms for tier 1.

Typing. The grouped keyboard was faster for both participants (Table 3), by 1.5 times for P1 and 2.8 times for P2, even though it needs almost two selections per letter. The dwell does not explain this: the mean dwell during the QWERTY runs was slightly shorter than during the grouped runs. Target size does. On the typing screen a letter group was 316 × 113 px and a QWERTY key 113 × 67 px, while the calibration error was 69 to 80 px, about the height of a QWERTY key. Figure 3 shows how widely P2's gaze spread around the targets on the compose screen. Speeds are low in absolute terms, as expected for first time users with a dwell above one second and no word suggestions.

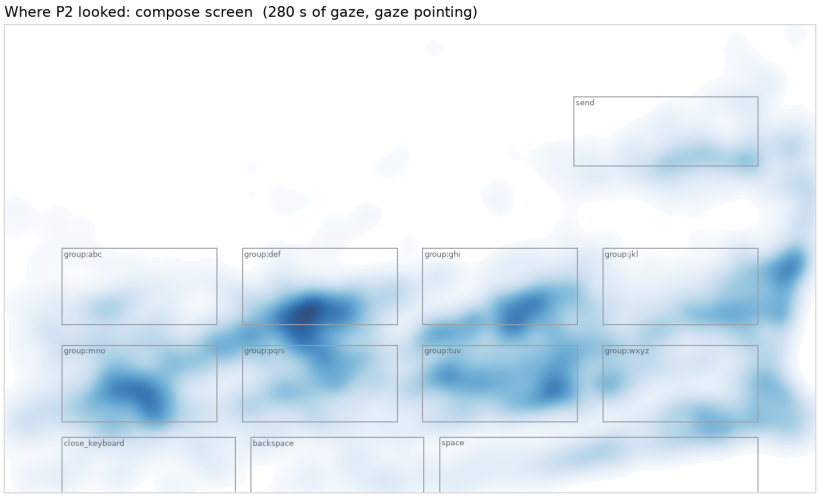


Fig. 3. Where P2 looked on the compose screen with the grouped keyboard, over the real button positions. Gaze spreads well beyond the targets, which favours few, large targets.

6 Discussion and Limitations

The early data support the design's main bet. Accidents stayed rare and confined to cheap, undoable actions, while the dwell on those actions came down by about a fifth within one session. No risky action fired by mistake, and the router began to drop the extra signal on tier 2 for a participant whose confirmations had worked. The typing results suggest that for gaze on a commodity phone, target size matters more than the number of selections per letter.

These results come from two able-bodied participants with no baseline condition, so they show that the learners behave as intended, not that adaptation beats a fixed dwell. Three limitations follow from the data. First, the typing runs were recorded while the dwell was still learning, and grouped runs always came first; we have since fixed the dwell during the typing test and will counterbalance the order. Second, our time of day rule treated a session after midnight as rested; the tired flag should use session length and time awake rather than the clock. Third, with a 1500 ms start and a 20 selection warm-up, the learner needs about 70 selections to settle, longer than a short study session.

Planned study. We are extending the study to two more able-bodied participants and one participant with a motor impairment. Each will use the app twice with gaze pointing, once with the adaptive learners and once with a fixed baseline (1000 ms dwell, dwell alone on tier 2, the preferred signal on tier 3), in counterbalanced order, with 5 minutes of practice, the four tasks, and the System Usability Scale [2] after each session. We will compare accidents and give-ups per 100 selections, task times and SUS with paired tests, and report the participant with a motor impairment as a case study.

7 Conclusion

Communic-aid shows that one consumer phone can provide both the pointing and the confirmation signals for a hands-free messaging interface, and that the interface can learn, from ordinary use and without asking, how long to wait and which signal to ask for before acting. Treating dwell and confirmation as two settings of one question, scaled by the cost of an error, gives a simple way to make gaze interaction both safer and faster for each person.

References

[1] Apple Inc. 2024. Apple Announces New Accessibility Features, Including Eye Tracking. https://www.apple.com/newsroom/2024/05/apple-announces-new-accessibility-features-including-eye-tracking/.

[2] John Brooke. 1996. SUS: A "Quick and Dirty" Usability Scale. In *Usability Evaluation in Industry*. Taylor & Francis, 189–194.

[3] Y. Cha, D. Chun, and S. Park. 2026. iPhoneme: Brain-to-Text Communication for ALS Using ConformerXL Decoding. arXiv:2604.16441

[4] Robert J. K. Jacob. 1990. What You Look at is What You Get: Eye Movement-Based Interaction Techniques. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '90)*. ACM, 11–18. doi:10.1145/97243.97246

[5] Lihong Li, Wei Chu, John Langford, and Robert E. Schapire. 2010. A Contextual-Bandit Approach to Personalized News Article Recommendation. In *Proceedings of the 19th International Conference on World Wide Web (WWW '10)*. ACM, 661–670. doi:10.1145/1772690.1772758

[6] Päivi Majaranta, Ulla-Kaija Ahola, and Oleg Špakov. 2009. Fast Gaze Typing with an Adjustable Dwell Time. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '09)*. ACM, 357–360. doi:10.1145/1518701.1518758

[7] M. Manzoni, Dragan Ahmetovic, and Sergio Mascetti. 2024. Personalized Facial Gesture Recognition for Accessible Mobile Gaming. In *Computers Helping People with Special Needs (ICCHP 2024)*. Springer, 120–127. doi:10.1007/978-3-031-62846-7_15

[8] Yogesh Kumar Meena, Hubert Cecotti, KongFatt Wong-Lin, and Girijesh Prasad. 2019. Design and Evaluation of a Time Adaptive Multimodal Virtual Keyboard. *Journal on Multimodal User Interfaces* (2019). doi:10.1007/s12193-019-00293-z

[9] Yogesh Kumar Meena and M. Salvi. 2025. Multimodal Appearance-Based Gaze-Controlled Virtual Keyboard with Synchronous-Asynchronous Interaction for Low-Resource Settings. doi:10.48550/arXiv.2508.16606

[10] Martez E. Mott, Shane Williams, Jacob O. Wobbrock, and Meredith Ringel Morris. 2017. Improving Dwell-Based Gaze Typing with Dynamic, Cascading Dwell Times. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems (CHI '17)*. ACM. doi:10.1145/3025453.3025517

[11] Omar Namnakani, Yasmeen Abdrabou, Jonathan Grizou, and Mohamed Khamis. 2025. GazeBlend: Exploring Paired Gaze-Based Input Techniques for Navigation and Selection Tasks on Mobile Devices. arXiv:2512.15491

[12] Anish S. Narkar, Jan J. Michalak, Candace E. Peacock, and Brendan David-John. 2024. GazeIntent: Adapting Dwell-Time Selection in VR Interaction with Real-Time Intent Modeling. arXiv:2404.13829

[13] Yesaya Tommy Paulus and Gerard Bastiaan Remijn. 2021. Usability of Various Dwell Times for Eye-Gaze-Based Object Selection with Eye Tracking. *Displays* 67 (2021), 101997. doi:10.1016/j.displa.2021.101997

[14] Björn R. Severitt, Nora Castner, Yannick Sauer, and Siegfried Wahl. 2026. A Real-Time Error Prevention System for Gaze-Based Interaction in Virtual Reality Based on Anomaly Detection. arXiv:2601.15146

[15] R. William Soukoreff and I. Scott MacKenzie. 2003. Metrics for Text Entry Research: An Evaluation of MSD and KSPC, and a New Unified Error Metric. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '03)*. ACM, 113–120. doi:10.1145/642611.642632

[16] William R. Thompson. 1933. On the Likelihood that One Unknown Probability Exceeds Another in View of the Evidence of Two Samples. *Biometrika* 25, 3/4 (1933), 285–294.

[17] WebAIM. 2013. Survey of Users with Motor Disabilities. https://webaim.org/projects/motordisabilitysurvey/.

[18] Jacob O. Wobbrock, Shaun K. Kane, Krzysztof Z. Gajos, Susumu Harada, and Jon Froehlich. 2011. Ability-Based Design: Concept, Principles and Examples. *ACM Transactions on Accessible Computing* 3, 3 (2011). doi:10.1145/1952383.1952384

[19] S. Zhang, Y. Gu, Kirsten Cater, and Oussama Metatla. 2026. Beyond Accuracy: Auditing Allocative Harms in Facial-Gesture Recognition for People with Motor Impairments. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26)*. ACM. doi:10.1145/3772318.3791927

[20] Xiaoyi Zhang, Harish Kulkarni, and Meredith Ringel Morris. 2017. Smartphone-Based Gaze Gesture Communication for People with Motor Disabilities. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems (CHI '17)*. ACM, 2878–2889. doi:10.1145/3025453.3025790